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Quality signaling and demand for renewable energy technology: Evidence from a randomized field experiment

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ABSTRACT

Solar technologies have been associated with private and social returns, but their technological potential often remains unachieved because of persistently low demand for high-quality products. In a randomized field experiment in Senegal, we assess the potential of three types of quality signaling to increase demand for high-quality solar lamps. We find no effect on demand when consumers are offered a money-back guarantee but increased demand with a third-party certification or warranty, consistent with the notion that consumers are uncertain about product durability rather than their utility. However, despite the higher willingness to pay, the prices they would pay are still well below market prices for the average household, suggesting that reducing information asymmetries alone is insufficient to encourage wider adoption. Surprisingly, we also find that the effective quality signals in our setting stimulate demand for low-quality products by creating product-class effects among those least familiar with the product.

1. Introduction

Many goods and services markets are characterized by imperfect and asymmetric information. One common problem is that low-quality products are observably indistinguishable from high-quality products at the point of purchase, leading to a "market for lemons" (Akerlof, 1970). The result is an inefficient outcome in which high-quality products are pushed out of the market since consumers that would otherwise be willing to purchase them are unwilling to pay the appropriate premium when their quality is unknown. This issue is especially germane in low-income countries, where institutions to ensure product quality are limited. Many theoretical and empirical studies have examined different quality-signaling approaches[1] to overcome these challenges (for literature reviews, see, for example, Riley, 2001; Dranove and Jin, 2010; Heiman and Hildebrandt, 2018). While a variety of quality signals have helped correct quality perceptions across diverse empirical settings, no studies compare the ability of different signaling interventions to address different aspects of the asymmetric information problem in one setting. Moreover, the evidence base for the effects of various signaling approaches on actual consumer demand is quite uneven[2]—particularly

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[1] Following quality-signaling theory, we use the term *quality signal* to encompass all potential mechanisms to overcome information asymmetries about product or service quality between buyers and sellers. Examples of quality-signaling mechanisms are the product price (for example, Wolinsky, 1983; Milgrom and Roberts, 1986), advertising (for example, Nelson, 1970; Schmalensee, 1982), warranties (for example, Spence, 1977; Grossman, 1981), and certification (for example, Auriol and Schilizzi, 2015; Chatterjee, 2020).

[2] The effects of third-party disclosure have been empirically examined in a range of settings including report cards for education and health services (for example, Dranove et al., 2003; Hastings and Weinstein, 2008), hygiene scores for restaurants (for example, Dai and Luca, 2020), certification for sports-card traders (Zhe Jin, Kato, and List, 2010), buyer-satisfaction scores for online sellers (Klein et al., 2016), certification of fertilizer quality (Michelson et al., 2021), and press releases about employers found to be violating safety and health standards in a recent inspection (Johnson, 2020).

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for quality assurance mechanisms.[3] We fill this void by running a randomized field experiment to compare the impacts of several popular and empirically understudied quality-signaling interventions on consumer demand for renewable energy technology in a market setting in sub-Saharan Africa.

Decentralized solar technologies are associated with significant private and social returns in developing countries.[4] A wide range of service levels is available, allowing households to dynamically scale and address varying electricity demands. Moreover, with grid electricity either not (yet) available, too expensive, or unreliable (for example, Lee et al., 2020; Aarakit et al., 2021; Deutschmann et al., 2021; Valickova and Elms, 2021), they have emerged as low-cost and clean alternatives—particularly in rural areas (Deichmann et al., 2011; Grimm et al., 2020). However, despite potential benefits and decreasing prices due to declining production costs (for example, Glemarec, 2012), households still appear to underinvest in solar technology in terms of scale and quality. Several market imperfections have been identified as potential barriers. Credit and liquidity constraints significantly inhibit households from purchasing high-quality solar technologies, but alleviating these constraints is currently insufficient to raise the willingness to pay to reach cost-covering prices (Abdul-Salam and Phimister, 2019; Grimm et al., 2020; Sievert and Steinbuks, 2020). In addition, providing accurate information during the sales process (Mahadevan et al., 2023) and information about the technologies' private and public benefits (Mekonnen et al., 2023) has not been found to significantly increase demand for high-quality products.

In this paper, we examine whether information asymmetry concerning product quality may contribute to the low willingness to pay for solar technology. Markets for off-grid solar products in sub-Saharan Africa are largely unregulated and dominated by highly affordable products without quality verification that may break easily, harming the environment through avoidable electronic waste (Kinally et al., 2022) and eroding confidence in the technology (Lighting Global, 2020).[5] While there are fixed and formal retail shops that sell quality-verified products (for example, Total's petrol stations), a large share of off-grid products are also marketed by itinerant vendors, who typically go from door-to-door selling products of unknown quality. In the latter case, information problems are likely to be exacerbated because such vendors have limited reputational capital at stake. While information asymmetries are frequently mentioned as a barrier to adoption (Chowdhury and Mourshed, 2016; Urpelainen, 2016; Stojanovski et al., 2017; Aklin et al., 2018; Balls, 2020; Groenewoudt et al., 2020; Samarakoon, 2020; Wassie and Adaramola, 2021), quality-disclosure and -assurance programs aimed to overcome the problem have not been analyzed.[6]

We contribute to the literature by providing the first empirical evidence on the effect of three quality-signaling mechanisms in Senegal on the demand for solar lamps[7]: third-party certification, warranties, and money-back guarantees (MBGs). These three treatments aim to either disclose or assure quality to potential consumers (Dranove and Jin, 2010). A certificate is defined as an effort to systematically measure and disseminate verifiable information about product quality. The quality is frequently disclosed to the consumer through a reputable independent third-party provider, which makes the information (more) credible (Auriol and Schilizzi, 2015; Stahl and Strausz, 2017; Zapechelnyuk, 2020). Warranties and guarantees convey information to the consumers about the quality of a product for a different reason: seller costs are increased if product quality is low (Spence, 1973). They are typically provided by the private sellers themselves and, thus, require product quality to be sufficiently high to be economically feasible (Kuksov and Lin, 2010; McWilliams, 2012). Warranties are designed to insure against product performance failure and durability risks (Spence, 1977; Gal-Or, 1989; Padmanabhan and Rao, 1993; Heiman and Hildebrandt, 2018), while MBGs are designed to insure against a poor fit between consumer preferences and product features by allowing consumers to experience the good for a certain amount of time (Moorthy and Srinivasan, 1995; Heiman et al., 2015; Heiman and Hildebrandt, 2018). By directly contrasting the effect of these different quality-signaling tools on actual consumer decision-making in a single experiment, we are able to distinguish between two key domains of asymmetric information driving the inefficiencies in markets for products: durability[8] and usefulness.

More specifically, our field experiment provides potential consumers in rural Senegal with the option to buy a high-quality solar lamp or a solar lamp of relatively low quality (to which we refer, for brevity, as a low-quality lamp). We hypothesize that the inability of households to differentiate between low- and high-quality lamps may lead to uncertainty about the lamps' durability and usefulness for the household once purchased. This uncertainty can lead consumers to avoid large investments in solar lamps even though, in expectation, relative to a low-

[3] There is little empirical evidence on quality signaling through assurance mechanisms, despite their prominent role in quality-signaling theory (for example, Spence, 1977; Grossman, 1981; Gal-Or, 1989). One example is provided by Roberts (2011), who analyzes the effect on overall market prices of the introduction of a market-wide buyer-satisfaction guarantee. Another example is Lewis (2011), who examines the effects of warranties on demand by comparing products for which the warranty period has ended with products with an active warranty.

[4] Evidence suggests that relative to other lighting sources such as battery-driven flashlights or kerosene-driven lamps, solar lamps substantially reduce household energy expenditures and improve educational outcomes for children (Bensch et al., 2012; Hassan and Lucchino, 2016; Grimm et al., 2017; Lemaire, 2018). Moreover, reducing consumption of dry-cell batteries plausibly leads to environmental benefits (Grimm et al., 2017; Stojanovski et al., 2017; Wagner et al., 2021), while reduced kerosene lamp usage has potential health benefits (Kudo et al., 2019; Lam et al., 2018).

[5] VeraSol (formerly Lighting Global Quality Assurance; https://verasol.org/about-quality-assurance/) is the world's most widely recognized quality-assurance framework for pico-solar technologies. Its webpage discloses laboratory test results and is mainly tailored to an audience of sector experts. While some products are designed and manufactured well, others fall short of expectations with regard to quality (that is, safety, durability, and performance). For instance, a test was conducted among a group of top-selling non-quality-verified solar products in five domestic markets across Africa and South Asia in 2017. The results showed that almost all products have one or more deficiencies affecting product durability (Lighting Global, 2018). Moreover, Groenewoudt et al. (2020) analyze the Ugandan off-grid solar product market and identify three market segments: high-quality, low-quality, and no-quality products, with the latter comprising junk and fake products. They conclude that of all market segments, the high-quality products will offer the most reliable, high-quality energy access.

[6] We acknowledge that the quality differences between quality-verified and unverified solar lighting products is currently somewhat debated in the literature (Girardeau et al., 2021). A few studies suggest that the quality differences are small. For instance, Bensch et al. (2018) explain that unverified solar products (more specifically solar home systems) in Burkina Faso have similar service levels compared to quality-verified systems and calculate that they incur lower costs under the assumption of a lifespan of at least two years. However, the authors also report a disadvantage of unverified solar products in terms of subjective user-satisfaction ratings.

[7] The lamps come with a solar panel that charges the battery by producing electrical energy from sunlight. They provide basic lighting and phone-charging capabilities, meeting the first level of energy access (Tier 1) defined by the World Bank's multi-tier framework (Bhatia and Angelou, 2015).

[8] There may also be differences in treatment effect sizes between the two durability signals. This can occur, for instance, when concerns arise about the credibility of the external party or the likelihood that the warranty will be honored. Additionally, the strength of the warranty signal is inversely related to the transaction costs of replacement, such as transportation to the retail shop. The importance of these transaction costs depends on the perceived value of the product. Hence, the relative effect size of the two interventions is ambiguous.

quality solar lamp, high-quality products bring lower lifetime costs at higher performance levels and therefore are likely to provide a higher return on investment.

Our randomized interventions are largely scalable. The third-party quality-assessment scheme was established by the World Bank Group's Lighting Africa Program with the explicit aim of overcoming purchase uncertainty and, by doing so, accelerating market growth of high-quality solar products to help achieve low-carbon energy-access targets (Murphy and Sharma, 2014; Lee et al., 2020). For MBGs and warranties to be effective, they require that people believe they will be honored, even if there is uncertainty in the quality of the product itself (Jeng et al., 2014). We partnered with Total Senegal to ensure warranty credibility, which would allow us to isolate the quality-signaling mechanism of the warranty and MBG. Trust in the organization,[9] and their wide network of fixed-location shops in the country relative to dedicated solar distributions firms that tend to have branches in major cities only, made them a natural partner. Economic agents distributing solar lamps in the market can aim to mimic our setup by joining or building up a warranty scheme through partnering with credible firms. Our analysis of the demand effects of such schemes allows us to shed light on the incentives for private sellers to invest in these types of quality-signaling tools in a lower-income-country setting, where it is much less common than in advanced economies.

The quality difference between the lamps used in the experiment is unobservable by visual inspection. Accordingly, households reveal no preference for one or the other at baseline. We measure demand for both products by eliciting the participants' willingness to pay (WTP) using the approach developed by Becker et al. (1964). This elicitation method is a real-purchase bidding game, which allows us to elicit revealed WTP. When assessing the demand for the high-quality lamp, we randomly assign the participants either to receive one of the three quality signals—third-party certification, a two-year warranty (2YW), or a six-week MBG—or to participate in the control group.

We find positive effects on WTP for high-quality lamps presented with the certificate or warranty. In contrast, we observe no effect on WTP for products offered with the MBG. These findings are consistent with the notions that the consumers' uncertainty is largely about the product's durability rather than its utility and that both quality disclosure and warranty are effective approaches for conveying information about product durability. However, while this lack of information can partially explain the low demand for high-quality solar lamps, signaling product durability only partially closes the gap between WTP and market prices. The substantial increase in WTP of about 12–14 percent observed for high-quality solar lamps presented with certification or the 2YW is in the range of effect sizes that have been observed for interventions aimed to ease credit and liquidity constraints (for example, Grimm et al., 2020). This implies that the impacts of the quality signals that were observed to be effective in our experiment also fall short of shifting the average WTP beyond the full market price. However, given recent VAT exemptions and reductions in import tariffs on quality-compliant products in Senegal (Lighting Global, 2022a), we estimate that the Lighting Global certification (LGC) and the 2YW have the potential to lift WTP above the new effective market price for around 15 percent of the sample. Thus, our quality signals are expected to affect demand for high-quality solar lamps under these new circumstances.

Surprisingly, we find that although the treatment only applies to the high-quality lamp, the certification and warranty treatments also increase WTP for the low-quality lamp—that is, a competing product of inferior quality with similar observable characteristics. Thus, quality signals appear to change perceptions about the entire product class and not solely the specific product at which they are targeted (Ackerberg, 2001). A similar phenomenon is also reported in the marketing literature (see for example, Anderson and Simester, 2013). For instance, Dubé and Manchanda (2005) report that advertising spending of one company not only increases demand for its own products but also boosts demand for the products of competitors. Similarly, more recent economics literature finds pharmaceutical television advertising to increase demand not only among the advertised drug, but also the non-advertised competitors in the same class (Shapiro, 2018).[10] These product-class effects reduce incentives of private sellers to invest in a warranty scheme, as they have to share the generated additional demand with competitors.

We probe this result by examining the role of consumer experience with solar lamps and find that the observed increase in WTP for the low-quality lamp is concentrated among inexperienced consumers.[11] Moreover, given that both the 2YW and the MBG are provided in cooperation with Total Senegal, but that we do not observe product-class effects for the MBG, we can conclude the significant results observed are due to the warranty itself, rather than to any reputational gain from the product's inclusion in the portfolio of a reputable company.

Taken together, our results suggest that quality-signaling treatments that convey information about the product's durability yield the classical prediction of a separation of low- and high-quality demand curves for experienced consumers (Akerlof, 1970), while the same interventions for inexperienced consumers appears to convey information about solar lamps more generally. It is also only among the former that we find a separation in the quality perceptions between the two lamps. Among inexperienced consumers, the quality-signaling treatments seem to serve as a general information campaign that drives up demand for both types of lamp.

Certification and (to some extent) warranties have broader applicability to markets in which product quality is variable and difficult to infer upon initial inspection, such as those for agricultural inputs, food, and repair services (Bai, 2021; Michelson et al., 2021). Our results seem most relevant for rural areas in low-income countries across the globe, especially where vendors go from one place to another selling products of unknown quality. By demonstrating the applicability of well-established economic theory to experienced consumers in a natural-experiment field setting (Harrison and List, 2004), our results strengthen the case for applying this theory to other market settings (List, 2020). However, our results also underscore important differences in the treatment effects between experienced and inexperienced consumers with welfare implications that represent an important area for future research.

The remainder of the paper is structured as follows. In Section 2 we introduce the study design, which includes a description of the solar lamps in the experiment, the quality-signaling treatments, and the method for eliciting WTP. Section 3 presents and discusses our empirical strategy. Section 4 shows the descriptive statistics. We provide our core results in Section 5 and discuss them in Section 6. Section 7 explores underlying mechanisms, and Section 8 concludes.

[9] In the context of focus group discussions with rural villages in proximity of our study sites, we inquired about Total's credibility and reputation at the design stage of our experiment. This is confirmed by our own survey data, where roughly 65 percent of the sample reports to trust Total Senegal (a similar percentage to trust in the World Bank, which is at roughly 72 percent). The sole role of Total Senegal was to serve as a partner for the warranty scheme and the MBG in our experiment.

[10] This strand of literature tends to refer to these as "spillovers" or "carryover effects" to describe this pattern of behavior. We prefer the term product-class effect to avoid confusion with other associations for those terms in economics.

[11] Various studies in the quality-signaling literature find heterogeneous consumer responses to third-party information based on consumer experience with the product (Jin and Sorensen, 2006; Wedig and Tai-Seale, 2002; Dutta et al., 2022) and consumer familiarity with the product (Dafny and Dranove, 2008). Sinkinson and Starc (2018) find that the effects of drug television advertising on the demand for drugs from the non-advertised competitors in the same class are mostly driven by new customers.

2. Study design and data

We conducted a randomized field experiment in the Fatick region of Senegal and elicited revealed WTP for low- and high-quality solar lamps. In October 2018, we visited 976 households in forty remote villages without access to grid electricity. Under the supervision of a field coordinator, eighteen enumerators implemented a door-to-door campaign[12] to run the WTP elicitation exercise for the solar lamps, provide the quality-signaling treatment for the high-quality solar lamp, and issue a short survey. The study was introduced to participants with the goal to learn about energy usage in rural Senegal to inform the World Bank's project work. Fig. 1 summarizes the study design. In each village, the surveyed compounds[13] were selected after a rapid assessment of the village population conducted with the help of traditional authorities (that is, village heads). All selected compounds willing to comply with the rules of the WTP elicitation exercise were eligible to participate. We selected and interviewed only one household per compound to limit information spillovers regarding the quality-signaling treatments.

When a household agreed to participate, enumerators began a survey with the household head to capture basic socioeconomic characteristics and energy consumption patterns. If the household head was not available, we interviewed the spouse of the household head. Therefore, our dataset contains information from one individual per household. A randomization procedure embedded in the electronic questionnaire then assigned respondents to (i) a lamp-presentation order (that is, low- or high-quality solar lamp first) and (ii) one of the three quality-signaling intervention groups. A fourth group was the control group, for which the high-quality solar lamp was offered without any quality-signaling intervention. The randomization was conducted on the individual level. In total, 976 participants took part in the survey and the subsequent elicitation exercise; 488 participants were presented with the low-quality lamp first, and the remaining 488 participants were shown the high-quality lamp first. Our final sample size comprises 960 households, because 16 households provided no WTP for either the low- or high-quality lamp. The different quality-signaling groups and the control group were roughly the same size (that is, one-fourth of the total). The treatment was delivered by the enumerator when eliciting WTP for the high-quality solar lamp. The enumerators explained to participants that the respective treatment only applied to the high-quality solar lamp but did not directly communicate information about its quality. Moreover, study participants were informed in advance that they would be asked to state their WTP for both lamps but that only one lamp would be offered to them at the end of the WTP elicitation exercise. The lamp offered was determined with a random draw. In the remainder of the section, we describe the two types of lamps and provide a short overview of the solar technology market in Senegal. We then present the three quality-signaling intervention treatments and briefly describe the bidding game for eliciting WTP.

2.1. Solar lamps offered in the experiment

For the high-quality solar lamp, the d.light s20 was procured from retail shops in Dakar and offered to experimental subjects. This product can illuminate a single room for four hours once fully charged and has a life expectancy of five years. It does not allow for phone charging. For reference, it costed approximately XOF 8,500 at the time of the study (approximately USD 15) when sold at Total Senegal shops located at high-traffic points along the road network. In our study areas, the lamp is not otherwise readily available at village markets. The s20 lamp was certified by the third-party organization Lighting Global after undergoing a series of laboratory tests to assess its quality on multiple dimensions.[14]

In comparison, lamps of lower quality are more commonly found in local markets; however, we procured the high- and low-quality lamps from the same retail shops in Dakar. The low-quality lamps are not sold at Total Senegal shops and costed approximately XOF 2,500 (around USD 4) at the time of the study, with a life expectancy of approximately six months. They are observationally similar to the high-quality lamp (see Fig. A1 in Online Appendix A), and similarly do not allow for phone charging. Among the control group—to which the high-quality lamp was not presented with any quality signal—the two lamps were indeed valued at similar prices.[15]

The rural electrification rate in Senegal is still under 65 percent (Gouvernement de la République du Sénégal, 2025), and where the grid is available, the service is characterized by low reliability and high prices (Signify Foundation, 2019). Yet rural households in Senegal have high demand for electricity—in fact, Deutschmann et al. (2021) find that both households and businesses are willing to pay a premium over current tariffs to receive quality electricity service without outages. Therefore, until reliable and affordable grid connections are available, off-grid solar technologies will offer a convenient option for meeting electricity needs in rural Senegal. However, quality standards are insufficiently enforced; only a few respectable solar distributors offer high-quality products and only a handful of retailers in rural Senegal invest in signaling quality. Recently (after the experiment was conducted), Senegal ramped up efforts toward creating a more favorable environment for high-quality solar companies. In 2020, Senegal joined other sub-Saharan African countries in providing VAT or duty exemptions for solar energy kits and components. Moreover, Senegal fully adopted the globally recognized International Electrotechnical Commission standards in January 2021, which contain quality standards for solar kits developed by the Lighting Global program (Lighting Global, 2022a).

2.2. Quality-signaling treatments

We tested and compared three interventions designed to address information asymmetries and risk associated with purchasing a new and potentially low-quality product: (1) third-party certification, (2) 2YW, and (3) MBG.

2.2.1. Third-party certification

In the certification intervention, an enumerator explained to the participant that the d.light s20 solar lamp's quality was certified by Lighting Global. The enumerator presented a certificate that verified the quality and included the Lighting Global and World Bank logos (see Fig. A2 in Online Appendix A). It provides very little information about the product itself or the specific tests that were conducted during the certification process—thus, the certificate is unlikely to shed light on the utility a consumer might experience from the product. While the LGC applies to all d.light s20 solar lamps, the certification information is only available on the Lighting Global website, meaning that few, if any, of the customers in our sample were previously aware of its existence. Lighting Global systematically measured different durability and performance features of the lamp, and it reports the lamp's quality on the certificate. Hence, while the certificate may not insure against durability and performance risk (Heiman and Hildebrandt, 2018), it serves as a quality signal by disclosing information (on the lamp's durability and performance) delivered by an independent third party (Dranove and Jin, 2010;

[12] Solar lamps are sold not only in conventional markets but through door-to-door campaigns in which salespeople market the lamps at consumers' homes.

[13] Compounds are well-defined groupings of typically two to five related households that share common areas within a clear geographical boundary.

[14] For more details on quality standards by Lighting Global, see https://www.lightingglobal.org.

[15] The difference in the WTP between the high-quality and low-quality solar lamps is not statistically significant conditional on having experience with the technology (see Table 4).

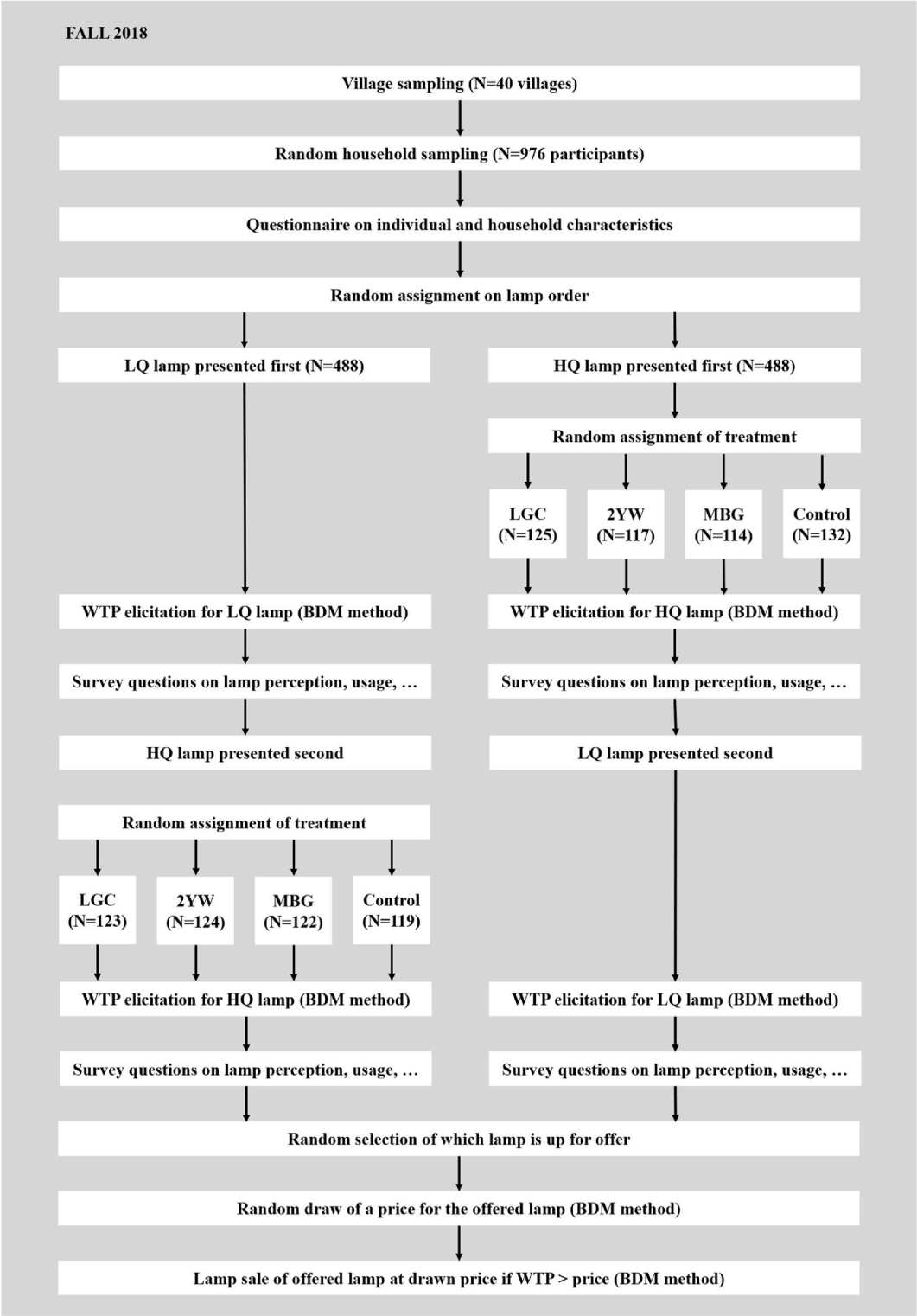

Fig. 1. Study design
Note: The final sample consists of 960 participants; 16 participants did not provide responses to either one of the WTP measures. LQ = low quality, HQ = high quality, LGC = Lighting Global certification, 2YW = two-year warranty, MBG = money-back guarantee, BDM method = Becker-DeGroot-Marschak method.

Auriol and Schilizzi, 2015).

2.2.2. *Two-year warranty*

The warranty intervention assured participants that the solar lamp comes with a 2YW that allows the participant to replace the lamp at Total Senegal shops if it breaks during the warranty period. The sole role of Total Senegal under this experimental condition was to serve as partner for the warranty scheme. Because of their significant network of shops, travel time from the sampled villages to these stations does not exceed an hour by car. Additionally, they are generally located near market centers that are regularly visited by our study participants for other market purchases and, thus, the marginal cost of a journey to

return a lamp is likely to be low since it could be bundled with regular household chores.

The product also comes with a two-year manufacturer warranty; however, in practice, this information is not salient, as customers do not have a viable way of contacting the manufacturer (in China) in case the lamp fails. For these reasons, Total Senegal had initiated its own warranty scheme for the product prior to the design phase of our experiment (see Fig. A3 in Online Appendix A). The warranty signals quality by assuring customers they will be aided in the event of mechanical failure and by reducing durability and performance risk (Spence, 1977; Heiman and Hildebrandt, 2018). This matters because these product features are often hard to assess at the time of purchase (Emons, 1989; Boulding and Kirmani, 1993; Padmanabhan and Rao, 1993).

2.2.3. Money-back guarantee

The MBG assures participants they will be able to return the product for a full refund at a Total Senegal shop within a six-week period if they are unhappy with it for any reason. For this intervention, we specifically partnered with Total Senegal and created the guarantee scheme for the study to test whether uncertainty is connected to the fit between the product and the consumer. In contrast to the 2YW, the short period of the MBG would not be sufficient for households to assess durability since a breakdown was unlikely to occur within six weeks. Hence, we anticipated that the MBG would be ineffective if durability was a primary concern. It is precisely for this reason that we view the MBG as insurance against a bad fit rather than bad quality, with the idea that participants could learn within a six-week period if the lamp proved sufficiently useful to the household post adoption. In choosing the length of the MBG, we followed the typical length of a thirty-to sixty-day period (Moorthy and Srinivasan, 1995). By offering a costly MBG and allowing the user to experience the product, the issuer of this guarantee signals high product and brand quality (Kuksov and Lin, 2010; McWilliams, 2012). Hence, the MBG reduces performance and fit uncertainties by allowing consumers to experience the good for a certain amount of time (Heiman et al., 2001; Heiman and Hildebrandt, 2018).

For both the warranty and MBG to work, the consumer has to perceive them as believable (Jeng et al., 2014); for this reason, the guarantee coupon carries both the World Bank Lighting Africa label and the Total Senegal logo (see Fig. A4 in Online Appendix A). The MBG is also unlikely to affect quality perception of the low-quality lamp as the average and median number of weeks households expect such low-quality lamps to last tends to significantly exceed this six-week window.[16]

2.3. Elicitation of willingness to pay

Our WTP elicitation process followed the Becker-DeGroot-Marschak (Becker et al., 1964), or BDM, elicitation method. The BDM exercise and the offer of the two lamps (including the presentation of the respective quality signal) were made by the enumerators, and not the partner company Total.

Respondents first participated in a practice round in which they were asked to make a price offer for two small bars of soap.[17] Hence, the practice round was designed to both demonstrate the random product draw and the random price draw of the assigned product. To explain the logic of the BDM procedure, they were told that a random price (within a range that approximates actual market prices) would be drawn and, if the price turned out to be higher than their offer, they would not be able to purchase the item. If the price turned out to be lower than their offer, they would be required to purchase the item at the drawn price. Besides the practice round, enumerators provided practical examples to illustrate the different scenarios under the BDM approach (that is, the consequences of bidding too low or too high relative to their indifference point) and allowed participants to ask questions on the procedure. After addressing all questions, the enumerator asked for the respondent's consent to participate in the BDM exercise, and confirmation that they had understood the procedure. When eliciting the WTP for each product, the enumerators did not provide a price range or price lists in order to avoid anchoring the participants bid to a specific level. Above and beyond the treatments, the enumerators provided only minimal product information. The enumerator script included a response to a predefined set of questions that were frequently asked by rural households in proximity to the study area at the pilot stage. These questions are around the product's features such as brightness levels, battery life, durability, materials, charging options, usage, origin, compatibility with charging phones, retail price, and purchase locations. However, this information was only provided if prompted by the participant.[18] Before locking in the final WTP, enumerators illustrated the potential outcome of their bid under different randomly drawn prices. Respondents had the possibility to adjust their bids or to confirm their initially stated WTP before proceeding with the WTP elicitation for the next lamp. Once the elicitation process for the WTP for the second lamp commenced, they could no longer revisit the WTP of the first lamp. They also had to commit to paying for the solar lamp on the day of the enumerator's visit. Participants did not receive any compensation for participating in the survey and the WTP elicitation exercise.

We selected the BDM process because it is an incentive-compatible approach for eliciting revealed WTP (Berry et al., 2020) and overcomes many of the biases associated with other forms of elicitation (for example, Blumenschein et al., 2008).[19] Each surveyor was required to follow a standardized script on their tablet and to never indicate the quality of the different lamps. The script included the BDM rules, a description of the lamps, and, when applicable, the details of the quality-signaling intervention. This helped to ensure that the offer was provided consistently across households and that participants judged the quality of the product on their own. We provide the script of the WTP elicitation exercise in Section F of the Online Appendix.

3. Empirical strategy

To estimate the impact of each intervention, we use the following specification:

$$logWTP_i = \alpha + \sum_{j=1}^{3} \beta_j T_{ji} + \epsilon_i \tag{1}$$

Here, WTP_i is the outcome of interest for participant i (namely, the WTP for the solar lamp in log form),[20] $j = \{1, 2, 3\}$ represents each of the three treatment arms, T_{ji} is a dummy variable equal to 1 if participant i is assigned to receive treatment j and 0 otherwise. β_j is the estimate of the average effect of intervention j. We estimate Equation (1) for both solar lamps separately and look at the difference in WTP between the high- and low-quality lamps as a third outcome to test whether there is a separation of demand curves.

[16] We gather this information from a different study in relation to renewable energy adoption in rural areas of two adjacent regions.

[17] In the practice round with soap, participants were not obliged to buy the soap.

[18] There is no information in our data indicating whether or not any questions were actually asked by the households.

[19] If the participants overstated their real reservation prices, they would have to buy the product at a price higher than their actual valuations. However, by understating their real reservation prices, they might miss a purchase opportunity at a price less than or equal to their valuations.

[20] Because of some large outliers, we winsorize the participants' WTP using a conservative cutoff at the two lower and upper percentiles.

4. Descriptive statistics

Table 1 summarizes the key sociodemographic characteristics of the participants; their trust levels toward the World Bank, Total Senegal, and nongovernmental organizations (NGOs) in general; and their past experience with solar lamps. The average monthly available income is XOF 42,758 across both groups. About 60 percent have a monthly available income higher than the XOF 8,500 market price for the high-quality solar lamp. Moreover, Table 1 shows that randomization into treatment and control groups was generally successful, with only one unbalanced characteristic: participants in the control group are slightly more willing to take risks (p-value = 0.077). The magnitude of this statistically significant difference is small. More detailed balance tests on each treatment and the control group are shown in Tables B1–B6 in Online Appendix B, and reveal that between each of the groups, there are further small significant differences in selected characteristics. For instance, there are marginally significant income differences between some experimental groups. However, these do not seem to follow systematic patterns that are consistent with the experimental findings presented below.[21] We estimate Equations (1) and (2) with and without (unbalanced) control variables to show that our findings are insensitive to any bias induced by the small imbalances between each treatment and the control group.

5. Results

The WTP for both the high- and low-quality lamp is positive for every person in the sample except one. Among the 960 participants in our survey, 515 placed a successful offer (that is, they stated a WTP equal to or above the drawn price) and were offered the lamp that was designated during the random draw. Of the 515 participants, 31 (or 6 percent of successful bidders) did not follow through with the purchase. The main reason was liquidity constraints coupled with overconfidence in being able to secure the money before the surveyors' departure.[22] The remaining 484 participants were able to pay for the offered solar lamp in full on the day the experiment took place. The high follow-through supports the argument that the BDM approach successfully elicits real demand.

Another convenient feature of the BDM method is that it allows us to trace a detailed demand curve by eliciting exact point-of-purchase prices. As our primary outcome of interest, the next subsection starts with the demand for the high-quality d.light s20 solar lamp measured through WTP elicited using the BDM approach.

5.1. Demand for the high-quality solar lamp

To visually portray the effect of each quality-signaling intervention on the demand for the high-quality solar lamp, Fig. 2 shows separate demand curves for all three treatment groups and the control group.

The blue line depicts the demand for the high-quality solar lamp when no quality signal is presented with it. While the MBG (orange line) does not lead to a significant increase in WTP, we observe an upward shift of the demand curves among participants who received the LGC or the 2YW. Thus, the treatments that reduce durability and performance risk either through disclosing information or through an insurance scheme lead to a statistically significant increase of XOF 303–42 (approximately USD 0.50).

We observe the same pattern when estimating Equation (1). The results are presented in Table 2. The effect of each treatment is indicated (with and without controls) in Columns (1) and (2). Since we observe a positive effect of similar magnitude for the LGC and the 2YW,[23] we pool these two treatments in Columns (3) and (4) and exclude participants who received the MBG. Observing no significant effect for the MBG but significant effects for the other two treatments gives credence to our experimental setup, as we would expect broader impacts in the presence of, for instance, experimenter demand effects or halo effects stemming from reputation gains resulting from the presentation of the logos of highly reputable institutions (the World Bank and Total Senegal). The World Bank logo appears on the LGC and the MBG, and the Total Senegal logo is imprinted on the 2YW and the MBG. Given that we observe significant positive effects for the LGC and the 2YW but not the MBG, neither the World Bank logo nor the Total Senegal logo seems to be driving our results.

On average, the WTP for the high-quality lamp increases by approximately 12–14 percent when the participant is presented with a quality signal about durability. This finding is consistent with the notion that the consumers' uncertainty is largely about product durability rather than product utility.[24] All effects of the durability signals on WTP for the high-quality lamp are significant when using monthly income as a covariate, when controlling for the WTP of a benchmark good (Dizon-Ross and Jayachandran, 2022), after adjusting for multiple hypothesis testing (List et al., 2019), when conducting randomization inference (Heβ, 2017), when measuring WTP in levels, and with different levels of winsorization (see Online Appendix C).

In terms of effect size, our findings are comparable to the results of other studies addressing barriers to solar technology adoption. Grimm et al. (2020) find that relaxing liquidity constraints increases WTP for different solar kits by at most 13 percent in Rwanda. Yoon et al. (2016) analyze whether a trial period and postponed payments increase WTP for solar lamps in India. They observe that the postponed payment increases WTP by about 17 percent. In line with our findings for the MBG, they find no evidence for the effectiveness of a trial period. Both studies conclude that the observed increases in WTP are not enough to reach cost-covering prices, as do we. With a market price of XOF 8,500, the average WTP of around XOF 2,700 when the high-quality solar lamp is presented with the LGC or the 2YW—though higher than when presented without any quality signal—is insufficient for the participants to purchase it at prevailing prices. Hence, while information asymmetries about product quality seem to be a relevant barrier to adoption, other barriers, such as credit, affordability, and liquidity constraints, factor in as well.

As mentioned in Section 2, VAT exemptions and reductions in import tariffs on quality-compliant products have effectively lowered prices for consumers since 2020 (Lighting Global, 2022a). Assuming a VAT of around 18 percent and import tariffs of around 23 percent, the market price of the high-quality lamp would be expected to fall to approximately XOF 5,000 (assuming pass-through of savings to consumers). Thus, while average WTP still remains below market prices, the LGC and the 2YW each have the potential to lift WTP above this new market price for around 15 percent of the sample (see Fig. 2).

[21] Moreover, when controlling for the income level rather than living in a house with a finished roof in our analyses, our main results remain qualitatively unchanged (Tables C1-C3, Online Appendix C).

[22] The thirty-one decliners were relatively equally distributed among the three treatment arms: nine decliners were in the control group, ten in the LGC group, four in the 2YW group, and eight in the MBG group. However, the thirty-one decliners are less equally distributed among the two types of lamps: twenty-eight were among those that were assigned to receive the high-quality lamp and three were among those that were assigned to receive the low-quality lamp. Yet, the allocation logarithm was overall skewed toward assigning the high-quality lamp (that is, 90 percent of the overall sample were assigned to the high-quality lamp).

[23] The null of the equality of the coefficients for the LGC and the 2YW in Columns (1) and (2) in Table 2 cannot be rejected at conventional levels.

[24] Corroborative evidence that concerns about usefulness of the device are indeed of less relevance in our setting is provided by the fact that no respondent requested the money back within six weeks.

Table 1
Balance test between control and treatment groups.

	Control (C)		Treatment (T)		C−T	
	(1)	(2)	(3)	(4)	(5)	(6)
	Mean	SD	Mean	SD	Diff.	p-value
Age (years)	47.96	15.46	49.05	15.24	−1.09	0.335
Female	0.32		0.33		−0.01	0.672
No education	0.62		0.59		0.03	0.377
Primary school	0.19		0.23		−0.04	0.231
Secondary school	0.12		0.10		0.02	0.474
Higher education	0.01		0.02		−0.01	0.432
Other: Koranic school/Daara	0.06		0.06		0.00	0.804
Finished roof[a]	0.64		0.69		−0.05	0.131
Household size (members)	12.80	7.25	13.07	7.19	−0.27	0.617
Monthly available income (XOF)[b]	37,459.87	52,798.25	44,613.70	82,711.12	−7,153.83	0.202
Share of income spend on traditional lighting[c]	0.06	0.15	0.06	0.16	0.00	0.870
Willingness to take risks (scale 0–10)[d]	4.95	3.50	4.49	3.55	0.46	0.077
Trust in World Bank[e]	0.70		0.72		−0.02	0.522
Trust in Total Senegal[e]	0.63		0.66		−0.03	0.450
Trust in NGOs[e]	0.60		0.64		−0.04	0.212
Solar lamp experience[f]	0.47		0.49		−0.02	0.498
Village-level solar lamp penetration rate[g]	0.49	0.18	0.48	0.18	0.01	0.334
Observations	249		711			

Note: The table presents baseline balance comparison using data collected during the household survey prior to making the solar lamp offer. Columns (1) and (3) report the mean values for the households assigned to control and treatment groups, respectively. All three treatments are combined in Column (3). Columns (2) and (4) report the respective standard deviations. Column (5) shows the difference between the control and treatment means and Column (6) the p-values for the test of equality of means. The joint null of equal means is not rejected at standard levels ($F(16, 940) = 0.920$, p-value = 0.541). [a] Binary variable taking the value of 1 if the main material of the roof is concrete, cement, tile or slate ("finished"), and 0 if zinc, thatch, straw, or clay ("rudimentary"). [b] The monthly available income consists of the sum of self-reported monthly available farm income, nonfarm income, remittances, benefits, wages, and rental income. [c] This includes the monthly expenditures for the following traditional lighting sources: kerosene, batteries, candles and generator. [d] Response to question "How do you see yourself in general: are you a person who is generally willing to take risks, or do you try to avoid taking risks? Please use a scale from 0 to 10, where 0 means you are 'completely unwilling to take risks' and a 10 means you are 'very willing to take risks.'" [e] Binary variable taking the value of 1 if the response to the statement "I trust the World Bank/Total/NGOs" was strongly agree or agree, and 0 if disagree, strongly disagree, or neither agree nor disagree. [f] Binary variable taking the value of 1 if the household owns a solar kit, and 0 otherwise. [g] The village-level solar lamp penetration rate is calculated using the average number of sampled households owning a solar kit within the surveyed households within each village. Hence, it is a village-level solar lamp penetration rate subject to the specific experimental sampling framework, offering a rough approximation of the actual share of households with past solar experience. All villages have at least one person reporting to have owned a solar lamp.

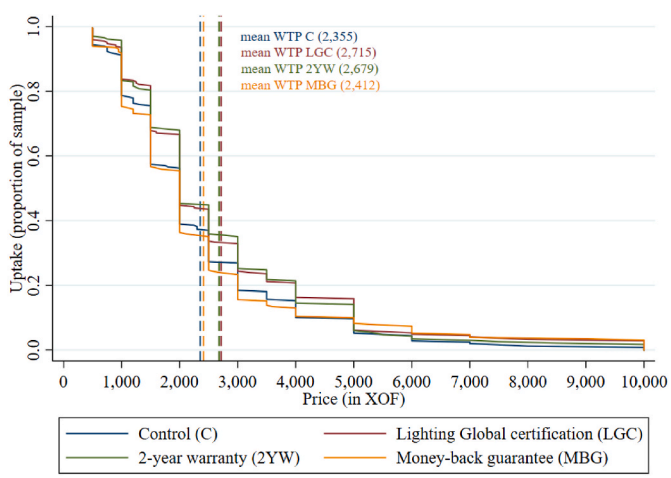

Fig. 2. Treatment effect on WTP for high-quality solar lamp
Note: WTP is winsorized at the two lower and upper percentiles. XOF 1,000 ≈ USD 1.60. Two-sample Kolmogorov-Smirnov test for equality of distribution functions provides (a) a p-value = 0.067 that control-group values are significantly smaller than LGC-treatment values; (b) a p-value = 0.036 that control-group values are significantly smaller than 2YG-treatment values; and (c) a p-value = 0.805 that control-group values are significantly smaller than MBG-treatment values.

Table 2
Treatment effects on WTP for high-quality solar lamp.

	(1)	(2)	(3)	(4)
	Log WTP for high-quality solar lamp			
LG certification	0.13**	0.14**		
	(0.06)	(0.06)		
2-year warranty	0.14**	0.12**		
	(0.06)	(0.06)		
Money-back guarantee	−0.01	−0.04		
	(0.06)	(0.06)		
Durability signaling (LGC, 2YW)			0.14***	0.12**
			(0.05)	(0.05)
Constant	7.56***	7.48***	7.56***	7.48***
	(0.04)	(0.13)	(0.04)	(0.14)
Control-group mean WTP	XOF 2,355.42			
Controls	NO	YES	NO	YES
Observations	960	958	729	727
R-squared	0.01	0.06	0.01	0.07

Note: WTP is winsorized at the two lower and upper percentiles. Robust standard errors are in parentheses. Individual and household controls are age, gender, education, finished roof, and household size. Risk and trust controls are willingness to take risks, trust in World Bank, trust in Total Senegal, and trust in NGOs. We furthermore control for solar kit experience, solar lamp penetration rate in the village, and the order of lamp presentation. The sample in Columns (3) and (4) consists of only the participants receiving the LGC and 2YW treatments and those in the control group. *** indicates significance at 1 %, ** at 5 %, and * at 10 %.

5.2. Demand for the low-quality solar lamp

Akerlof's (1970) model predicts a separation of the demand curves for low- and high-quality products when the latter are presented with signals that allow customers to distinguish their quality. In our experiment, we indeed observe a small but positive shift of the demand curve for high-quality lamps with the LGC and 2YW. To assess the impact of

quality signaling for high-quality lamps on the demand for low-quality lamps, we further elicited the WTP for a relevant competing product of inferior quality.

We estimate Equation (1) with revealed WTP for low-quality lamps as an outcome; the results are presented in Table 3. While we hypothesized that the quality-signaling treatments would have a negative or null effect on WTP for low-quality lamps, we instead observe that the interventions significantly *increase* demand for these lamps. On average, participants receiving either the LGC or 2YW for the high-quality lamp increase their WTP for the low-quality lamp by approximately 10–11 percent.[25] On the contrary, the MBG has no statistically significant effect on the respective WTP. The Online Appendix C shows that the effects of the durability signals on WTP for the low-quality lamp remain significant when using monthly income as covariate, when controlling for the WTP for a benchmark good (Dizon-Ross and Jayachandran, 2022), after adjusting for multiple hypothesis testing (List et al., 2019), when conducting randomization inference (Heβ, 2017), and with different levels of winsorization. Moreover, the effects are concentrated among participants randomly presented with the high-quality lamp first (see Table D1 in Online Appendix D).[26] This would be consistent with participants making inferences from the quality signaling interventions onto the low-quality lamps shown later.

One possible concern is that the association with Total as a credible agency may be driving the effects on low- and high-quality lamps, rather than the quality signaling interventions being tested. The statistically insignificant effects of the MBG on WTP for the low-quality lamp suggests that consumers do not necessarily automatically assume high quality upon learning that a reputable company markets a similar product. Both the 2YW and the MBG carry the Total logo on the certificate—hence, the partnership with Total per se is unlikely to explain these effects. We conclude that these quality-signaling treatments appear to change perceptions about the entire product class and not solely the specific product they are attached to. In the rest of this paper, we refer to this finding as a "product-class effect", which we now explore further by examining the role of consumer experience with solar lamps.

Table 3
Effects of treatments for high-quality lamp on WTP for low-quality solar lamp.

	(1)	(2)	(3)	(4)
	Log WTP for low-quality solar lamp			
LG certification	0.09	0.11*		
	(0.06)	(0.06)		
2-year warranty	0.12**	0.10*		
	(0.06)	(0.06)		
Money-back guarantee	−0.01	−0.03		
	(0.06)	(0.06)		
Durability signaling (LGC, 2YW)			0.10**	0.10**
			(0.05)	(0.05)
Constant	7.44***	7.49***	7.44***	7.52***
	(0.04)	(0.12)	(0.04)	(0.14)
Control-group mean WTP	XOF 2,090.16			
Controls	NO	YES	NO	YES
Observations	960	958	729	727
R-squared	0.01	0.05	0.01	0.05

Note: WTP is winsorized at the two lower and upper percentiles. Robust standard errors are in parentheses. Individual and household controls are age, gender, education, finished roof, and household size. Risk and trust controls are willingness to take risks, trust in World Bank, trust in Total Senegal, and trust in NGOs. We furthermore control for solar kit experience, solar lamp penetration rate in the village, and the order of lamp presentation. The sample in Columns (3) and (4) consists of only the participants receiving the LGC and 2YW treatments and those in the control group. *** indicates significance at 1 %, ** at 5 %, and * at 10 %.

6. Role of consumer experience

Various studies in the quality-signaling literature find heterogeneous consumer responses to quality signals based on experience with a good or service (Wedig and Tai-Seale, 2002; Jin and Sorensen, 2006; Dutta et al., 2022) and familiarity with it (Dafny and Dranove, 2008). Such findings have been frequently reported in the marketing literature (for example, Dubé and Manchanda, 2005). Differentiating between high- and low-quality products is already challenging in a market in which consumers are fairly familiar with the product class, but it could be exacerbated when the entire product class is relatively new to potential consumers. In this case, quality-signaling interventions might convey information not only about one product's quality but about the entire product class. In an explorative analysis, we empirically test this by differentiating effects between experienced and inexperienced customers.

With more than half of the survey participants having no solar kit experience (see Table 1), we can consider solar lamps to be a relatively new product class to this subsample of consumers and consequently a little-known alternative to commonly used lighting sources such as candles, battery-run flashlights, and kerosene lamps. Among inexperienced consumers, the randomized interventions referring to the high-quality lamp may convey information about solar lamps more generally, turning them into general information campaigns that (a) put solar lamps on the map as a viable lighting source and (b) highlight certain quality dimensions that traditional lighting sources do not perform well on.

To assess these new hypotheses, we estimate Equation (1) separately for participants with and without prior solar lamp experience.[27] We consider the same two outcomes as before plus one additional outcome: (a) log WTP for the high-quality lamp, (b) log WTP for the low-quality lamp, and (c) the difference in log WTP between the high- and low-quality lamps. As the two durability-signaling treatments again have similar effects (see Table D3 in Online Appendix D),[28] we pool the LGC and 2YW treatments and omit the sample receiving the MBG. As indicated in Columns (3) and (4) of Panel B in Table 4, the product-class effects are solely driven by participants without prior solar lamp

[25] The effects of both the LGC and 2YW are again similar in size, because the null of the equality of the coefficients for the LGC and the 2YW in Columns (1) and (2) in Table 3 cannot be rejected at conventional levels.

[26] In Table D2 of Online Appendix D, we also display the corresponding effects on the WTP for the high-quality lamp. As expected, lamp order is irrelevant for the effects of the warranty. For the certificate, we see a borderline difference for the warranty between households being presented with the offer of the high-quality lamp first and those who received it second (significant at the ten percent level, see Column [2]). Taking stock of these results: of the six plausibility checks leveraging the randomized lamp order (lamp order interaction with treatment for the certification, warranty and MBG, both for the low- and high-quality lamp), five of the six support the original findings, with the sixth having borderline significance where a null effect was expected. Given the multiple different tests conducted with disaggregated sub-groups, we interpret the results holistically as being in support of the presence of a product-class effect in the case of the warranty and certificate. For reassurance, we follow a rule-of-thumb multiple-hypothesis-testing adjustment, which indeed renders the previously borderline statistically significant test result statistically insignificant at conventional levels. In contrast, applying the same simplified adjustment to the p-values for the other three plausibility tests, the interaction terms remain significant at the ten percent level for the warranty and certificate. We use this simplified adjustment approach because the standard multiple-hypothesis-testing correction commands employed in the paper are not applicable to interaction terms.

[27] See Table B7 in Online Appendix B for a balance test between respondents with and without solar lamp experience. Solar lamp experience correlates with a variable indicating that the home has a finished roof and a variable capturing the education level of the respondent.

[28] When testing for the equality of coefficients, the null again cannot be rejected at conventional levels.

Table 4
Treatment effects on WTP, by level of experience.

	Full sample		No experience		Solar lamp experience	
	(1)	(2)	(3)	(4)	(5)	(6)
(A) Log WTP for high-quality solar lamp						
Durability signaling (LGC, 2YW)	0.17**	0.16**	0.17**	0.17**	0.10	0.09
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.08)
Solar lamp experience	0.03	0.05				
	(0.08)	(0.08)				
Treatment × experience	−0.07	−0.08				
	(0.10)	(0.10)				
Constant	7.55***	7.46***	7.55***	7.37***	7.58***	7.58***
	(0.05)	(0.14)	(0.05)	(0.18)	(0.06)	(0.24)
Observations	729	727	382	381	347	346
R-squared	0.01	0.07	0.02	0.08	0.01	0.09
(B) Log WTP for low-quality solar lamp						
Durability signaling (LGC, 2YW)	0.21***	0.21***	0.21***	0.22***	−0.01	−0.02
	(0.06)	(0.07)	(0.06)	(0.07)	(0.08)	(0.08)
Solar lamp experience	0.12	0.17**				
	(0.08)	(0.08)				
Treatment × experience	−0.22**	−0.23**				
	(0.10)	(0.10)				
Constant	7.39***	7.45***	7.39***	7.35***	7.51***	7.71***
	(0.05)	(0.14)	(0.05)	(0.20)	(0.06)	(0.22)
Observations	729	727	382	381	347	346
R-squared	0.01	0.05	0.03	0.07	0.00	0.08
(C) Difference in log WTP: high-quality minus low-quality solar lamp						
Durability signaling (LGC, 2YW)	−0.04	−0.05	−0.04	−0.05	0.11**	0.11*
	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)	(0.06)
Solar lamp experience	−0.09	−0.12**				
	(0.06)	(0.06)				
Treatment × experience	0.15**	0.16**				
	(0.07)	(0.07)				
Constant	0.16***	0.00	0.16***	0.03	0.07	−0.13
	(0.04)	(0.10)	(0.04)	(0.13)	(0.05)	(0.17)
Observations	729	727	382	381	347	346
R-squared	0.01	0.04	0.00	0.05	0.01	0.06
Controls	NO	YES	NO	YES	NO	YES

Note: WTP is winsorized at the two lower and upper percentiles. Robust standard errors are in parentheses. The sample consists of only the participants receiving the LGC and 2YW treatments (="Durability signaling") and those in the control group. Individual and household controls are age, gender, education, finished roof, and household size. Risk and trust controls are willingness to take risks, trust in World Bank, trust in Total Senegal, and trust in NGOs. We furthermore control for solar lamp penetration rate in the village and the order of lamp presentation. *** indicates significance at 1 %, ** at 5 %, and * at 10 %.

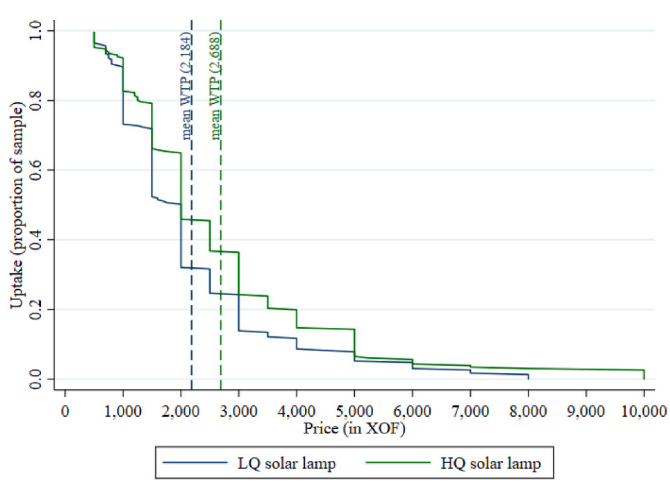

Fig. 3. Demand curves for low-quality and high-quality solar lamps among experienced participants
Note: WTP is winsorized at the two lower and upper percentiles. XOF 1,000 ≈ USD 1.60. Data are from the treatment group (LGC and 2YW) and experienced consumers.

experience. In the experiment, the inexperienced participants are introduced to a new product class. Without having any benchmark, they seem to increase their WTP for both the high- and low-quality lamps—considering either of them as a practical alternative to other lighting sources. A similar pattern has been observed elsewhere. For instance, Sinkinson and Starc (2018) suggest that the impact of pharmaceutical television advertising on the demand for products of non-advertised competitors within the same category is predominantly concentrated among new customers.

As presented in Columns (5) and (6) of Panels A and B in Table 4, we detect a small positive effect on WTP for high-quality lamps and a slightly negative effect on WTP for low-quality lamps among the experienced participants, though both effects are insignificant. To assess whether the latter effect is strong enough to induce a separation of the demand curves, we turn to Columns (5) and (6) in Panel C. The displayed coefficients indicate that the LGC and 2YW treatments lead to a significant separation of demand curves among participants with prior solar lamp experience. All effects presented in Table 4 remain significant when controlling for monthly income and WTP of a benchmark good (Dizon-Ross and Jayachandran, 2022), after adjusting for multiple hypothesis testing (List et al., 2019), when conducting randomization inference (Heβ, 2017), when measuring WTP in levels, and with different levels of winsorization (see Online Appendix C). To mitigate concerns that our estimated effects may be driven by certain enumerators presenting the quality signals as if they applied to both products, we

Table 5
Treatment effects on quality perception, by level of experience.

	Full sample		No experience		Solar lamp experience	
	(1)	(2)	(3)	(4)	(5)	(6)
(A) Good/excellent expected quality for high-quality solar lamp						
Durability signaling (LGC, 2YW)	0.04	0.04	0.04	0.03	0.08*	0.09*
	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)	(0.05)
Solar lamp experience	−0.05	−0.04				
	(0.05)	(0.05)				
Treatment × experience	0.04	0.03				
	(0.06)	(0.06)				
Constant	0.83***	0.75***	0.83***	0.61***	0.77***	0.89***
	(0.03)	(0.09)	(0.03)	(0.12)	(0.04)	(0.13)
Observations	691	689	358	357	333	332
R-squared	0.01	0.04	0.00	0.10	0.01	0.07
(B) Good/excellent expected quality for low-quality solar lamp						
Durability signaling (LGC, 2YW)	0.04	0.03	0.04	0.02	−0.05	−0.05
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Solar lamp experience	−0.01	−0.01				
	(0.06)	(0.06)				
Treatment × experience	−0.08	−0.08				
	(0.07)	(0.07)				
Constant	0.77***	0.67***	0.77***	0.57***	0.76***	0.81***
	(0.04)	(0.10)	(0.04)	(0.13)	(0.04)	(0.15)
Observations	678	676	350	349	328	327
R-squared	0.01	0.06	0.00	0.09	0.00	0.08
(C) Difference in good/excellent expected quality: high-quality minus low-quality solar lamp						
Durability signaling (LGC, 2YW)	0.01	0.03	0.01	0.03	0.12**	0.13**
	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)	(0.06)
Solar lamp experience	−0.03	−0.01				
	(0.06)	(0.06)				
Treatment × experience	0.10	0.09				
	(0.08)	(0.07)				
Constant	0.05	0.15	0.05	0.07	0.02	0.16
	(0.04)	(0.12)	(0.04)	(0.16)	(0.05)	(0.19)
Observations	669	667	347	346	322	321
R-squared	0.01	0.04	0.00	0.04	0.01	0.09
Controls	NO	YES	NO	YES	NO	YES

Note: Quality expectation is a binary variable taking the value 1 if the respondent expects good or excellent quality and 0 if the respondent expects very poor, poor, or acceptable quality. Robust standard errors are in parentheses. The sample consists of only the participants receiving the LGC and 2YW treatments (="Durability signaling") and those in the control group. Individual and household controls are age, gender, education, finished roof, and household size. Risk and trust controls are willingness to take risks, trust in World Bank, trust in Total Senegal, and trust in NGOs. We furthermore control for solar lamp penetration rate in the village and the order of lamp presentation. Overall, the sample sizes are slightly smaller than in Table 4 because some respondents refused to answer the perceived-quality question. *** indicates significance at 1 %, ** at 5 %, and * at 10 %.

show that they remain qualitatively unchanged when including enumerator fixed effects (Table D4 in Online Appendix D). To account for the possibility that inexperienced households may have never owned a solar lamp but were still exposed to them due to high village-level solar lamp penetration, we additionally divided the sample of households without solar lamp experience into two groups. One group consists of inexperienced households living in villages with below-median solar lamp penetration rates, and the other includes those in villages with above-median penetration rates. We find that the product-class effects are driven by the former group (see Table D5 in Online Appendix D), which further confirms that the observed product-class effects are driven by inexperienced consumers.

Fig. 3 visualizes this separation by graphing the two demand curves for the low- and high-quality lamps among experienced participants. Both treatments based on signaling durability induce a WTP gap of around XOF 500 (approximately USD 0.80).

7. Mechanisms: quality-perception updating

While we observe a separation of the demand curves for low- and high-quality solar lamps among experienced participants in line with Akerlof's (1970) prediction, we see a positive shift in demand for both solar lamps among inexperienced ones. We next further investigate these different responses to quality signals by turning to the underlying mechanisms.

After eliciting the WTP, participants were asked to rate the quality of each lamp on a five-point Likert scale: 1 (very poor), 2 (poor), 3 (acceptable), 4 (good), or 5 (excellent). To examine whether the treatments convey product-quality information to the consumers that leads to increased WTP, we use this variable as an outcome and estimate the following specification:

$$EQ_i = \alpha + \beta_j T_{ji} + \epsilon_i \quad (1)$$

Here, EQ_i is a binary measure of the expected quality of the lamp; it takes on the value of 1 if participant i rates the respective lamp to have good or excellent quality and 0 if participant i rates the respective lamp

to only have acceptable, poor, or very poor quality.[29] As before, we pool the LGC and 2YW treatments (indicated by T_{ji}) and omit the sample receiving the MBG. Table 5 presents the results of estimating Equation (2) for experienced and inexperienced participants separately; Panel A indicates the effects on perceived quality for the high-quality lamp and Panel B for the low-quality lamp, while Panel C indicates the effects on the difference in quality perceptions between the two lamps.

We begin by noting that the constant term is large, positive, and statistically significant. This may reflect cultural norms whereby respondents feel compelled to make positive statements in response to direct questioning about a particular experience, person, or object (DeMaio, 1984). It is also possible that our initial description of the survey's objective, which was to inform the portfolio of World Bank's development investments in rural Senegal, may have served as a quality signal to respondents. Besides raising overall quality perceptions, this may reduce the effect that the genuine quality signaling approaches may have on quality perceptions. Moreover, given the ex post quality assessment, respondents may state quality perceptions to justify their revealed WTP. As a result, the variation with which to identify the impacts of the treatment on quality perception is quite limited, constraining our ability to fully explore the mechanisms that may be driving our demand effects. Nonetheless, Columns (5) and (6) show that participants with prior solar lamp experience who received the LGC or 2YW treatment exhibit a marginally significant eight percentage-point higher likelihood of rating the offered high-quality lamp as good or excellent (see Panel A). Though the result is insignificant, they are also less likely to rate the low-quality lamp as good or excellent (see Panel B), thereby generating a significant separation in quality perception (see Panel C).

The results are less pronounced among participants without prior solar lamp experience (see Columns [3] and [4] in Table 5). The durability-signaling treatments lead to positive but insignificant increases in good and excellent ratings for the high-quality lamp but similar effects (albeit somewhat smaller after adding controls in Column [4]) for the low-quality lamp. Hence, compared with the strong WTP impacts among this group, the respective results for quality perception appear smaller. One explanation for this results pattern is that, relative to participants with prior solar lamp experience, those without experience display a much smaller sensitivity to perceived quality differences.[30] However, the point estimates for participants without prior solar lamp experience are supportive of a product-class effect that is consistent with our earlier results, yielding no significant separation of quality perceptions among inexperienced consumers.

8. Conclusion

We examined the effect of three quality-signaling interventions on the demand for high-quality solar lamps. The solar lamp market in rural Senegal—as in other rural areas of sub-Saharan Africa—exhibits patterns that are consistent with the common asymmetric information problem, in which consumers are unable to distinguish between low- and high-quality products at the point of purchase, leading to underinvestment in high-quality lamps despite their greater cost-effectiveness. In a context of unreliable grid infrastructure with low coverage, off-grid solar technologies offer a sustainable option for meeting electricity demand—especially in rural Senegal.

Our study illustrates that quality signals that either credibly inform consumers about a product's durability, such as third-party certification, or assure durability through a (two-year) warranty significantly increase demand for high-quality lamps. Among inexperienced consumers, we observed product-class effects, in which the two significant quality-signaling interventions induce an increased demand for not only high-quality lamps but also low-quality lamps, to which the quality signals are not attached. For this group of consumers, the interventions seem to function as a general information campaign, driving up the demand for the entire product class, irrespective of quality. Such product-class effects have been observed in advertising, where, for instance, television advertisement positively affects the demand for non-advertised competitor products (for example, Dubé and Manchanda, 2005; Shapiro, 2018). In a study on drug advertising, Sinkinson and Starc (2018) find that such effects are primarily driven by new customers, consistent with our findings. Only among experienced consumers do the quality-signaling treatments yield the classical prediction of a separation of low- and high-quality demand curves (Akerlof, 1970). The observed separation is induced by an overall increase in perceived quality of high-quality solar lamps conveyed through the two above-mentioned quality-signaling interventions.

However, these effects might not be enough to translate into actual purchases by the average household, because the difference between the WTP for the high-quality solar lamps and their prevailing market price is substantial. Among experienced consumers, for instance, the average WTP (approximately XOF 2,700) for high-quality solar lamps among respondents receiving a quality signal is still less than one-third of the market price (XOF 8,500). This suggests that additional relevant barriers to adoption persist, such as credit and liquidity constraints (for example, Abdul-Salam and Phimister, 2019; Grimm et al., 2020; Sievert and Steinbuks, 2020). Studies that examine the effect of relaxing these constraints in rural Senegal and India find similar effect sizes to those reported in our study, at around 13–17 percent, which are not enough to reach cost-covering prices in their settings either. The remaining affordability barrier may likely require subsidization schemes. According to Grimm et al. (2020), subsidies for solar technologies can be justified as policy instruments from a social planner's perspective to achieve universal access to energy and alleviate poverty. Hence, to generate meaningful demand for higher-quality solar lighting products among poor rural households and to address environmental risks posed by the growing volume of electronic waste from (low quality) solar products (Kinally et al., 2022), a combination of subsidies, credit, and quality signaling may be needed.

Additionally, declining production costs of solar panels and LED lights (Glemarec, 2012; Apostoleris et al., 2019; Valickova and Elms, 2021) and favorable policies for high-quality-lamp manufacturers such as VAT exemptions and changes in import tariffs are lowering prices for the consumer (Lighting Global, 2022b). In our setting, we calculated a reduction of the market price of the high-quality lamp from XOF 8,500 to 5,000 since the experiment was conducted. At this lower price, the LGC and the 2YW each have the potential to lift the WTP of 15 percent of the sample beyond the critical threshold.

The effectiveness and scalability of the third-party certificate and (to some extent) the warranty potentially give the interventions appeal for retailers in a wide range of markets in which product quality is variable and difficult to infer upon initial inspection. Third-party certification is likely to be the dominant intervention among those tested here. Although it does not insure against early breakdown, it can be provided

[29] Participants judged the quality based on the physical product shown by the enumerator, without being forced to provide a response if uncertain. Roughly five percent of the sample did not provide any response. The share of "don't know" responses is fairly evenly distributed among participants with and without prior experience with the product. When assuming a low-quality perception for "don't know" responses, that is, inputting a zero value among households without any response, we find qualitatively similar results as those presented in Table 5 (see Online Appendix E, Table E1).

[30] Estimates of the correlation between the WTP and our measure of quality perception for both inexperienced and experienced participant groups are available upon request.

at relatively low cost to both the supplier and the buyer.[31] Our finding of significant effects of the warranty suggests that households believed Total Senegal would honor the warranty even though the lamps in the respective treatment group were sold by a different party. However, as the offers were made by enumerators rather than actual itinerant vendors, results are transferable only to the extent that consumers continue to trust the authenticity of the warranty regardless of the vendor. To enhance trust in the warranty's authenticity, several measures may be helpful in this regard, such as adding a QR code to the warranty certificate that directs consumers to a reputable company's website, confirming its validity. The warranty, in theory, offers more comprehensive coverage of quality uncertainty by providing insurance against possible breakdown but comes at an additional cost to the supplier by potentially inducing moral hazard (that is, more risky usage; see, for example, Lutz and Padmanabhan, 1998) and to the buyer since they bear a transaction cost in traveling to the supplier to request an exchange. Additionally, given the low increase in WTP in relation to the market price, warranties might not be an economically viable option for the suppliers because they require risky and labor-intensive efforts. Moreover, even if vendors are inclined to offer a warranty, it remains uncertain whether credible firms would extend their warranty schemes to these vendors. Furthermore, it is unclear whether our experimental findings would materialize if the vendors themselves offered the warranty.

When testing generalizations from a widely used economic theory in a relevant market setting in which it should hold, we find that the theoretical predictions are valid for experienced customers only. This is important for two main reasons. First, it suggests that future theoretical research on quality signaling should develop extensions to nascent product markets. Second, from the perspective of suppliers, the product-class effects among inexperienced consumers are concerning. If a product class is novel or knowledge about it is not broadly disseminated, private firms will have little incentive to invest in quality-disclosure and quality-assurance efforts. In these situations, public interventions may be welfare enhancing but only if any potential welfare losses associated with the boost in demand for low-quality products is more than offset by gains due to increased sales of high-quality ones. The conditions under which product-class effects of this sort exacerbate or ameliorate the lemon problem are likely context specific and represent an important area for future research.

CRediT authorship contribution statement

Aidan Coville: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Joshua Graff Zivin:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Arndt Reichert:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ann-Kristin Reitmann:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization.

AER trial registry

This RCT was registered as AEARCTR-0002632 (Reichert and Coville, 2019). The analysis was not prespecified.

[31] This might not apply to all third-party certification services. Yet in the case of the Lighting Global certificate, the service is free and the only cost that may fall on the retailer is the one-time cost of changing the lamps' packaging to include information on the certification.

IRB approval

IRB approval for this study was obtained from IRB Solutions (Protocol #2016/03/4, 2018–08–17).

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jdeveco.2025.103514.

Data availability

Data will be made available on request.

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